

Course code:ITA0608

Course name:Machine learning

8. Write a program to implement Linear Regression (LR) algorithm in python.

Aim:

To implement the **Linear Regression (LR)** algorithm in Python and predict the output values based on the input data.

Algorithm:

1. Import the required Python libraries.
2. Load the dataset.
3. Select the input feature (X) and target (y).
4. Split the dataset into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict the output for the test data.
8. Calculate and display the model accuracy (R^2 score).

Program Code:

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import
train_test_split from sklearn.linear_model

import LinearRegression # Load dataset

data = load_diabetes()

X =

data.data y
```

```
= data.target
```

```
# Split datas X_train, X_test, y_train, y_test =
```

```
train_test_split( X, y, test_size=0.2,
```

```
random_state=42
```

```
)
```

```
# Create Linear Regression model
```

```
model = LinearRegression()
```

```
# Train the model
```

```
model.fit(X_train,
```

```
y_train) # Predict
```

```
y_pred =
```

```
model.predict(X_test) #
```

```
Accuracy (R2 Score)
```

```
score = model.score(X_test,
```

```
y_test) print("Predicted Values:")
```

```
print(y_pred[:10])
```

```
print("R2 Score:", score)
```

OUTPUT:

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load dataset
data = load_diabetes()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy (R2 Score)
score = model.score(X_test, y_test)

print("Predicted Values:")
print(y_pred[:10])
print("R2 Score:", score)
```

```
... Predicted Values:
[139.5475584  179.51720835 134.03875572 291.41702925 123.78965872
 92.1723465  258.23238899 181.33732057 90.22411311 108.63375858]
R2 Score: 0.4526027629719195
```

RESULT:

The **Linear Regression algorithm** was successfully implemented in Python. The model was trained on the dataset, predicted the output values, and its performance was evaluated using the **R² score**

